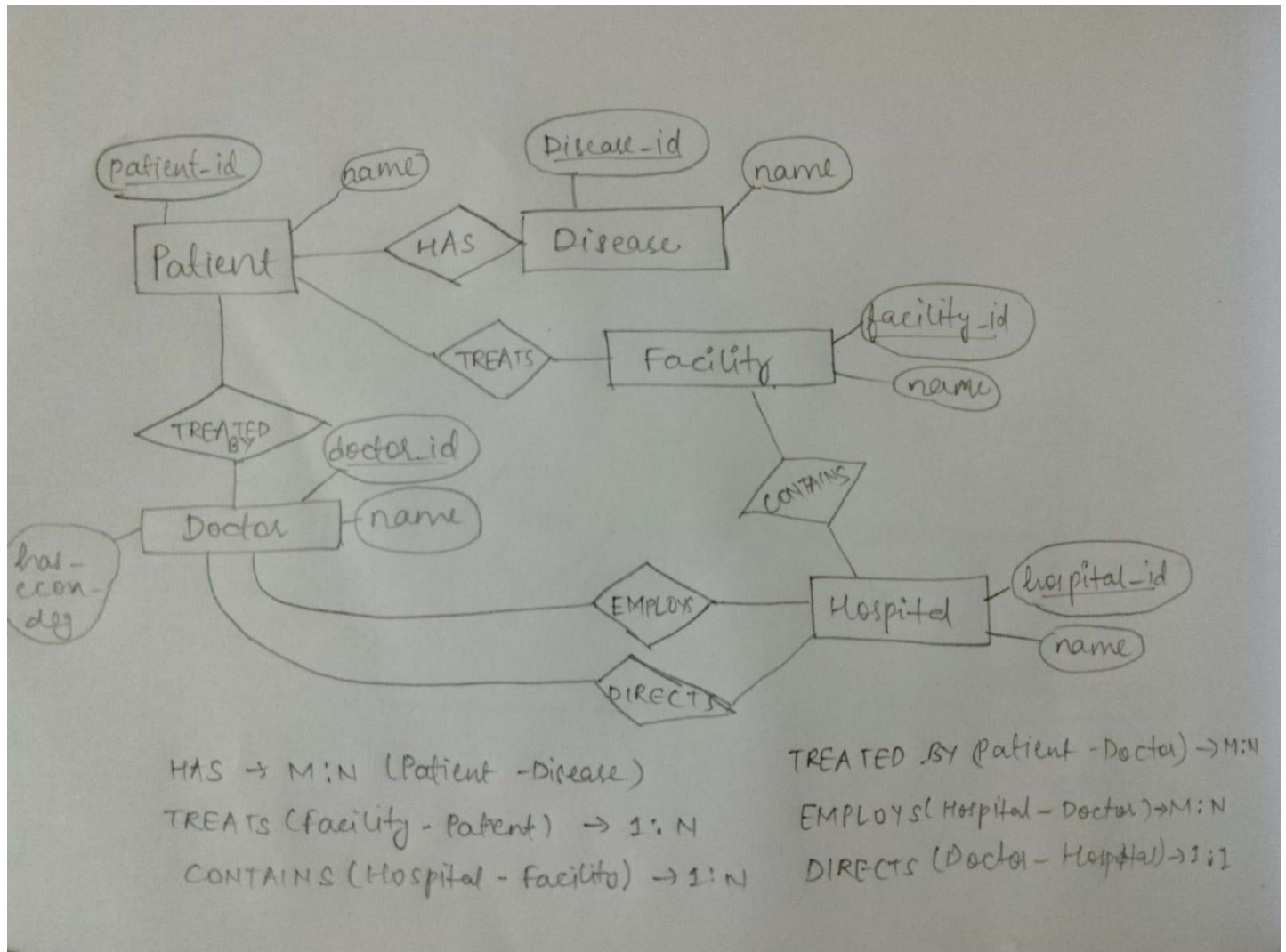


ASSIGNMENT-2

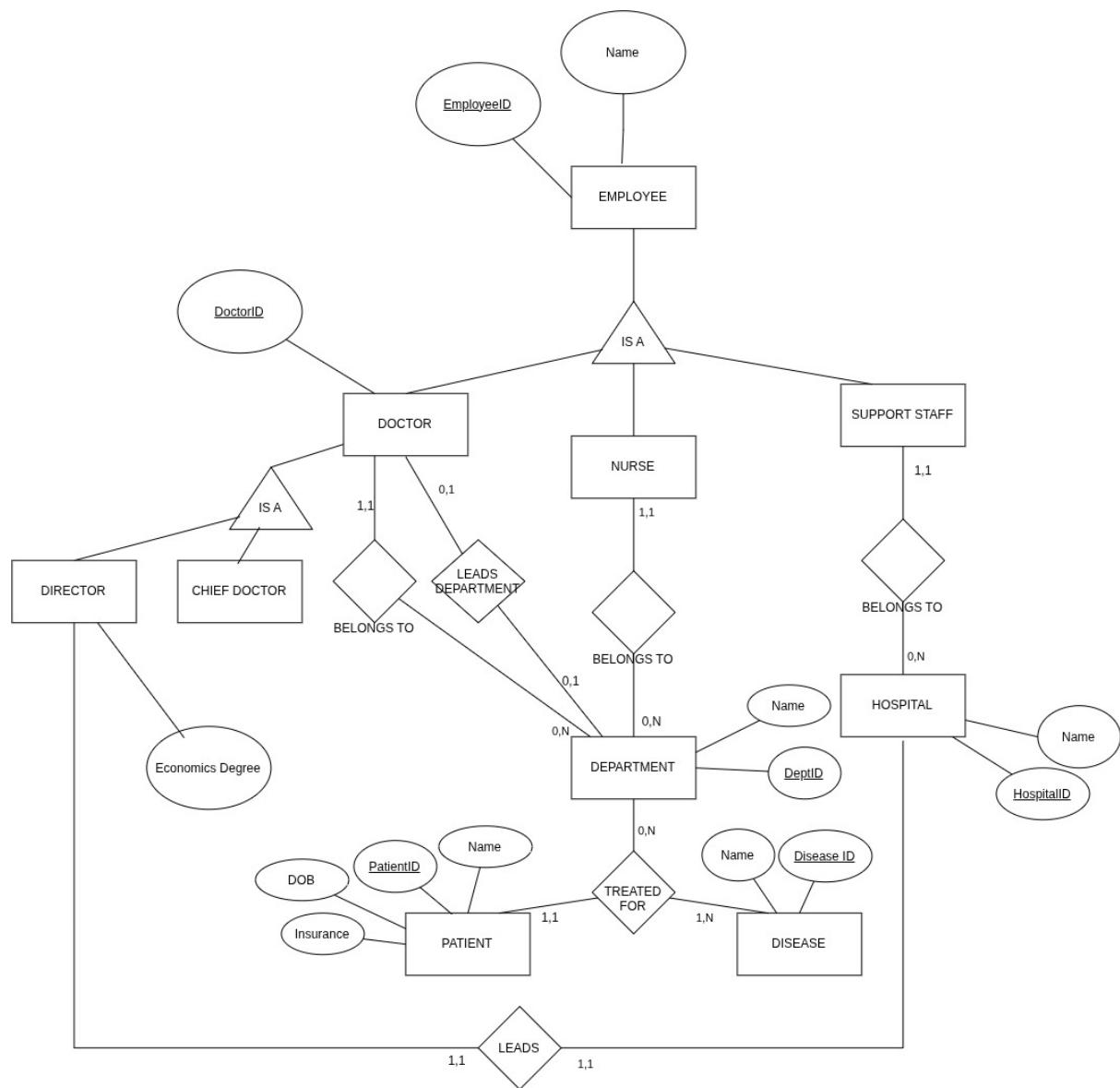
Q.1-

Correct ER Diagram :



Q.2-

Correct ER Diagram :



Q.3-

1. A(A1, A2, B1)

Primary Key: A1

Foreign Key: B1 references B(B1)

2.B(B1, B2, C1)

Primary Key: B1

Foreign Key: C1 references C(C1)

3. C(C1, C2, D1)

Primary Key: C1

Foreign Key: D1 references D(D1)

4. D(D1, D2, D3, A1T, A1S, A1U)

Primary Key: D1

Foreign Keys:

A1T references A(A1) — relationship T

A1S references A(A1) — relationship S

A1U references A(A1) — relationship U

Q.4-

1.a-

Entity Relations

1. PATIENT(patient_id, name)

PK: patient_id

2. DISEASE(disease_id, name)

PK: disease_id

3. FACILITY(facility_id, name)

PK: facility_id

4. DOCTOR(doctor_id, name, specialty)

PK: doctor_id

5. HOSPITAL(hospital_id, name)

PK: hospital_id

Relationship Relations

1. HAS(patient_id, disease_id)

PK: (patient_id, disease_id)

FK: patient_id → PATIENT(patient_id)

FK: disease_id → DISEASE(disease_id)

2. TREATS(facility_id, patient_id)

PK: (facility_id, patient_id)

FK: facility_id → FACILITY(facility_id)

FK: patient_id → PATIENT(patient_id)

3. CONTAINS(hospital_id, facility_id)

PK: (hospital_id, facility_id)

FK: hospital_id → HOSPITAL(hospital_id)

FK: facility_id → FACILITY(facility_id)

4. TREATED_BY(patient_id, doctor_id)

PK: (patient_id, doctor_id)

FK: patient_id → PATIENT(patient_id)

FK: doctor_id → DOCTOR(doctor_id)

5. EMPLOYS(hospital_id, doctor_id)

PK: (hospital_id, doctor_id)

FK: hospital_id → HOSPITAL(hospital_id)

FK: doctor_id → DOCTOR(doctor_id)

6. DIRECTS(doctor_id, hospital_id)

PK: doctor_id

FK: doctor_id → DOCTOR(doctor_id)

FK: hospital_id → HOSPITAL(hospital_id)

1.b-

1. PATIENT(patient_id, name, facility_id)

PK: patient_id

FK: facility_id → FACILITY(facility_id)

2. DISEASE(disease_id, name)

PK: disease_id

FK: None

3. FACILITY(facility_id, name, hospital_id)

PK: facility_id

FK: hospital_id → HOSPITAL(hospital_id)

4. DOCTOR(doctor_id, name, specialty)

PK: doctor_id

FK: None

5. HOSPITAL(hospital_id, name, doctor_id)

PK: hospital_id

FK: doctor_id → DOCTOR(doctor_id)

6. HAS(patient_id, disease_id)

PK: (patient_id, disease_id)

FKs:

patient_id → PATIENT(patient_id)

disease_id → DISEASE(disease_id)

7. TREATED_BY(patient_id, doctor_id)

PK: (patient_id, doctor_id)

FKs:

patient_id → PATIENT(patient_id)

doctor_id → DOCTOR(doctor_id)

8. EMPLOYEES(hospital_id, doctor_id)

PK: (hospital_id, doctor_id)

FKs:

hospital_id → HOSPITAL(hospital_id)

doctor_id → DOCTOR(doctor_id)

2.a-

Entity relations

1. EMPLOYEE(EmployeeID, Name)

PK: EmployeeID

2. DOCTOR(DoctorID, EmployeeID)

PK: DoctorID

FK: EmployeeID → EMPLOYEE(EmployeeID)

3. NURSE(EmployeeID)

PK: EmployeeID

FK: EmployeeID → EMPLOYEE(EmployeeID)

4. SUPPORT_STAFF(EmployeeID)

PK: EmployeeID

FK: EmployeeID → EMPLOYEE(EmployeeID)

5. DIRECTOR(DoctorID, EconomicsDegree)

PK: DoctorID

FK: DoctorID → DOCTOR(DoctorID)

6. CHIEF_DOCTOR(DoctorID)

PK: DoctorID

FK: DoctorID → DOCTOR(DoctorID)

7. DEPARTMENT(DeptID, Name)

1PK: DeptID

8. HOSPITAL(HospitalID, Name)

PK: HospitalID

9. PATIENT(PatientID, Name, DOB, Insurance)

PK: PatientID

10.DISEASE(DiseaseID, Name)

PK: DiseaseID

Relationship relations

1. BELONGS_TO — Doctor–Department

DOCTOR_BELONGS_TO(DoctorID, DeptID)

PK: DoctorID

FKs:

DoctorID → DOCTOR(DoctorID)

DeptID → DEPARTMENT(DeptID)

2. LEADS_DEPARTMENT — Doctor–Department

LEADS_DEPARTMENT(DoctorID, DeptID)

PK: DoctorID

FKs:

DoctorID → DOCTOR(DoctorID)

DeptID → DEPARTMENT(DeptID)

3. BELONGS_TO — Nurse–Department

NURSE_BELONGS_TO(EmployeeID, DeptID)

PK: EmployeeID

FKs:

EmployeeID → NURSE(EmployeeID)

DeptID → DEPARTMENT(DeptID)

4. BELONGS_TO — Support Staff–Hospital

STAFF_BELONGS_TO(EmployeeID, HospitalID)

PK: EmployeeID

FKs:

EmployeeID → SUPPORT_STAFF(EmployeeID)

HospitalID → HOSPITAL(HospitalID)

5. LEADS — Director–Hospital

LEADS(DoctorID, HospitalID)

PK: DoctorID

FKs:

DoctorID → DIRECTOR(DoctorID)

HospitalID → HOSPITAL(HospitalID)

6. TREATED_FOR — ternary relationship

TREATED_FOR(PatientID, DeptID, DiseaseID)

PK: PatientID

FKs:

PatientID → PATIENT(PatientID)

DeptID → DEPARTMENT(DeptID)

DiseaseID → DISEASE(DiseaseID)

2.b-

1. EMPLOYEE

EMPLOYEE(EmployeeID, Name)

PK: EmployeeID

2. DOCTOR

DOCTOR(DoctorID, EmployeeID, DeptID)

PK: DoctorID

FKs:

EmployeeID → EMPLOYEE(EmployeeID)

DeptID → DEPARTMENT(DeptID)

3. NURSE

NURSE(EmployeeID, DeptID)

PK: EmployeeID

FKs:

EmployeeID → EMPLOYEE(EmployeeID)

DeptID → DEPARTMENT(DeptID)

4. SUPPORT_STAFF

SUPPORT_STAFF(EmployeeID, HospitalID)

PK: EmployeeID

FKs:

EmployeeID → EMPLOYEE(EmployeeID)

HospitalID → HOSPITAL(HospitalID)

5. DIRECTOR

DIRECTOR(DoctorID, EconomicsDegree, HospitalID)

PK: DoctorID

FKs:

DoctorID → DOCTOR(DoctorID)

HospitalID → HOSPITAL(HospitalID)

6. CHIEF_DOCTOR

CHIEF_DOCTOR(DoctorID)

PK: DoctorID

FK: DoctorID → DOCTOR(DoctorID)

7. DEPARTMENT

DEPARTMENT(DeptID, Name, LeaderDoctorID)

PK: DeptID

FK: LeaderDoctorID → DOCTOR(DoctorID)

8. HOSPITAL

HOSPITAL(HospitalID, Name)

PK: HospitalID

9. PATIENT

PATIENT(PatientID, Name, DOB, Insurance)

PK: PatientID

10. DISEASE

DISEASE(DiseaseID, Name)

PK: DiseaseID

11. TREATED_FOR

TREATED_FOR(PatientID, DeptID, DiseaseID)

PK: PatientID

FKs:

PatientID → PATIENT(PatientID)

DeptID → DEPARTMENT(DeptID)

DiseaseID → DISEASE(DiseaseID)

Q.5-

1. E1(A11, A12, A13, R5_E3)

Key: A11

A11 → NOT NULL

A12 → NOT NULL

A13 → NOT NULL

R5_E3 → NOT NULL

Foreign key:

$R5_E3 \rightarrow E3(A31)$ (R5 has E1 (1,1), so every E1 must be related to E3.)

2. E2 (A21, A22, R1_E1)

Key: A21

$A21 \rightarrow \text{NOT NULL}$

$A22 \rightarrow \text{NOT NULL}$

$R1_E1 \rightarrow \text{NOT NULL}$

Foreign key:

$R1_E1 \rightarrow E1(A11)$ (E2 has (1,1) participation in R1.)

3. E3 (A31, R4_E1, AR4)

Key: A31

$A31 \rightarrow \text{NOT NULL}$

$R4_E1 \rightarrow \text{NULL allowed}$

E3 has (0,N) participation in R4, so an E3 may not participate in R4.

A AR4 is also nullable because it belongs to R4, and if an E3 has no R4 relationship, there is no AR4 value $R4 \rightarrow \text{NULL allowed}$

Foreign key:

$R4_E1 \rightarrow E1(A11)$

4. E4(A41, R2_E2, R3_E3)

Key: A41

$A41 \rightarrow \text{NOT NULL}$

$R2_E2 \rightarrow \text{NOT NULL}$

$R3_E3 \rightarrow \text{NOT NULL}$

Both are NOT NULL because E4 has (1,1) participation in both R2 and R3.

Foreign keys:

$R2_E2 \rightarrow E2(A21)$

$R3_E3 \rightarrow E3(A31)$